

NETWORK FUNDAMENTALS

HTB Academy · Module 2

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Your complete notes: Networks · OSI Model · TCP/IP · Protocols · Transmission

What is a Network?



A network is a collection of interconnected devices that can **communicate, send/receive data, and share resources.**



Real-World Analogy

Think of friends chatting in a room:



Each person = a **Node** (device)



Talking = **Communication Link**



Conversation = **Data being shared**



Nodes

Individual devices connected
(PC, phone, server, printer)



Links

Communication pathways
(wired or wireless)



Data Sharing

Primary purpose:
exchange of information

Why Are Networks Important?



Resource Sharing

Share hardware (printers)
&
software across devices



Communication

Email, instant messaging,
video calls — all via
networks



Data Access

Access files & databases
from any connected device



Collaboration

Work together in real-time
even miles apart

Networks have radically transformed society since the advent of the Internet 🌐

Types of Networks



Local Area Network

 **Scope**

Small area — home, school, office

 **Speed**

Very HIGH data transfer rates

 **Media**

Ethernet cables or Wi-Fi

 **Ownership**

Single person / organization

 **Example**

Your home Wi-Fi router

VS



Wide Area Network

 **Scope**

Cities, countries, continents

 **Speed**

Slower vs LAN (long distance)

 **Media**

Fiber optics, satellite, telecom

 **Ownership**

Multiple ISPs / organizations

 **Example**

The Internet itself

How LAN + WAN Work Together



🔧 **Modem Role:** Converts digital signals from your router → format suitable for transmission over telephone lines, cable, or fiber optics. It bridges your **LAN to the ISP's WAN**, giving all home devices internet access.

Basic Network Terminology

IP Address

Unique number for each device (like a home address)
192.168.1.1

DNS

Domain Name System: converts google.com → IP address

MAC Address

Permanent hardware ID by manufacturer
00:1A:2B:3C:4D:5E

DHCP

Auto-assigns IP addresses to devices. Saves manual setup

Router

Connects LAN to WAN (internet). Decides where data should go

Bandwidth

Amount of data per second. Higher = faster transfer

Switch

Connects devices inside a LAN. Sends data only to correct device

Latency

Delay in data transfer. Lower latency = faster response

The OSI Model — 7 Layers

7	Application	Network services to apps. HTTP, FTP, SMTP, DNS
6	Presentation	Data format, encryption, compression
5	Session	Manages sessions between apps. APIs
4	Transport	End-to-end delivery. TCP & UDP
3	Network	Logical addressing + routing. Routers, IP
2	Data Link	Node-to-node transfer. Switches, MAC addresses
1	Physical	Raw bit transmission. Cables, hubs, Ethernet

OSI Layers Deep Dive — Lower Layers (1-4)

L1 — Physical Layer

Transmits RAW BITS over physical medium
Hardware: Ethernet cables, hubs, repeaters
Deals with: voltage, timing, data rates
 Real life: The actual wire carrying electricity

L2 — Data Link Layer

Node-to-node data transfer between physically connected nodes
Devices: Switches, Bridges
Uses MAC addresses to identify devices
 Real life: A walkie-talkie between two rooms

L3 — Network Layer

Packet forwarding + routing across networks
Devices: Routers
Uses IP addresses for logical addressing
 Real life: GPS navigation deciding best route

L4 — Transport Layer

End-to-end communication for apps
Segmentation, reassembly, flow control
TCP = reliable | UDP = fast, no guarantee
 Real life: Courier ensuring package arrives intact

OSI Layers Deep Dive — Upper Layers (5-7)



L5 — Session Layer

Establishes, maintains, terminates connections (sessions)

Session checkpointing — resume after interruption

APIs coordinate communication between systems

🌐 Real life: A phone call — dialing, talking, hanging up



L6 — Presentation Layer

Translator between app and network format

Data encryption/decryption happens HERE

Data compression + format conversion

🌐 Real life: Google Translate between two people



L7 — Application Layer

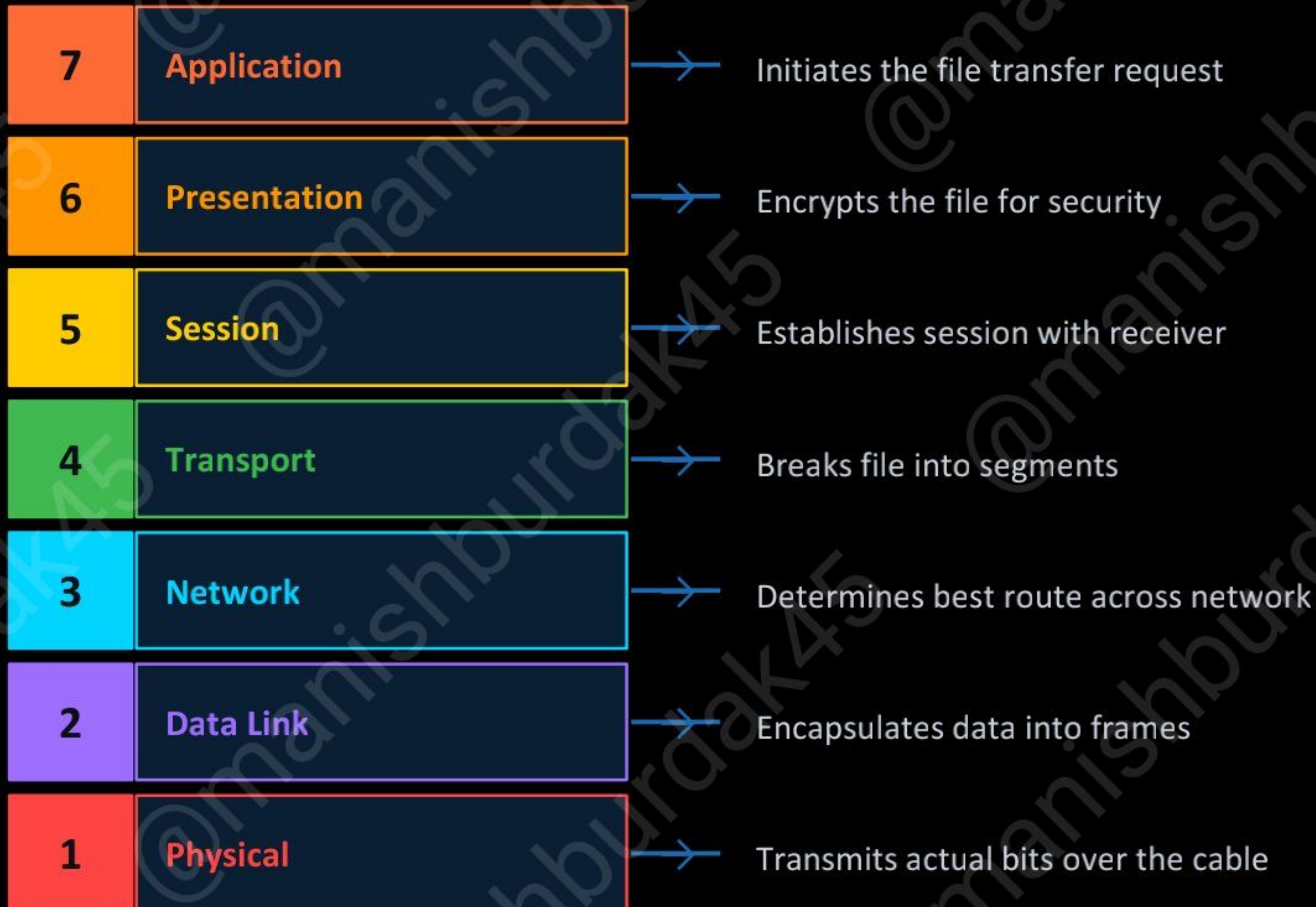
Directly interfaces with user applications

HTTP — web browsing | FTP — file transfer

SMTP — email | DNS — domain name resolution

🌐 Real life: The browser/app you actually use

Sending a File — OSI Layer by Layer



TCP/IP Model — 4 Layers



L4 — Application Layer

= OSI Session + Presentation + Application

HTTP, FTP, SMTP — actual services users interact with



L3 — Transport Layer

= OSI Transport

TCP (reliable) or UDP (fast). End-to-end delivery



L2 — Internet Layer

= OSI Network

IP + ICMP. Logical addressing and routing of packets



L1 — Link Layer

= OSI Physical + Data Link

Ethernet, Wi-Fi. Physical connection and data framing



Accessing a Website: Browser (App Layer) → TCP (Transport) → IP routing (Internet) → Physical wire (Link)

OSI vs TCP/IP — Side by Side

Aspect	OSI Model	TCP/IP Model
Layers	7 layers	4 layers
Purpose	Conceptual / Theoretical	Practical Implementation
Usage	Framework for understanding	Backbone of the Internet
Created by	ISO (International)	U.S. DoD / ARPANET
Flexibility	Very detailed, strict	Simpler, application-focused
Session layer	Separate (L5)	Merged into Application
Presentation	Separate (L6)	Merged into Application

PROTOCOLS

Common



IP

Net L3

Routes packets across networks. Logical addressing



HTTP

App L7

Transfers web pages between browser and server



FTP

App L7

Uploads/downloads files between systems



SMTP

App L7

Sends emails from one server to another



DNS

App L7

Resolves domain names to IP addresses



TCP

Trans L4

Reliable, connection-oriented. Error checking & recovery



UDP

Trans L4

Fast, connectionless. Ideal for streaming/gaming

TCP vs UDP — Transport Layer Protocols

✓ TCP

Transmission Control Protocol

- 🔗 Connection-oriented (handshake first)
- ✓ Guaranteed delivery — error checking
- 📦 Data arrives in correct ORDER
- 🐢 Slower — more overhead
- 📌 Use for: Web, Email, File Transfer
- 🌐 Like: Registered post — delivery confirmed

VS

⚡ UDP

User Datagram Protocol

- 🔓 Connectionless (no handshake)
- ✗ No guaranteed delivery
- 🔄 Order NOT guaranteed
- 🚀 Fast — minimal overhead
- 📌 Use for: Streaming, Gaming, VoIP
- 🌐 Like: Shouting in a crowd — fast, no ack

Transmission in Networking



Transmission Types

Analog: Continuous signals (radio, old telephone)

Digital: Discrete bits — modern computer networks, Wi-Fi



Transmission Modes

Simplex: One-way only (keyboard → PC)

Half-Duplex: Two-way, not simultaneous (walkie-talkie)

Full-Duplex: Both ways at once (phone call)



Transmission Media



Wired Media



Twisted Pair

Ethernet, LAN connections



Coaxial

Cable TV, early Ethernet



Fiber Optic

High-speed internet backbones



Wireless Media



Radio Waves

Wi-Fi, cellular networks



Microwaves

Satellite communications



Infrared

Short-range (remote controls)

Practice Questions — Set 1

Q1. What layer of the OSI model is responsible for physical connections like Ethernet cables?
(Format: two words)

Q2. Name the OSI layer that deals with logical addressing and routing.
(Format: two words)

Q3. Which protocol ensures reliable delivery of data at the Transport Layer?

Q4. At what layer do switches operate within the OSI model?
(Format: three words)

Q5. What layer of the TCP/IP model corresponds to OSI's Application, Presentation, and Session layers?
(Format: two words)

Practice Questions — Set 2

Q6. Which layer of the OSI model manages data encryption and format conversion?
(Format: two words)

Q7. Name a protocol used for web browsing that operates at the Application Layer.

Q8. Which OSI layer ensures segments are transferred reliably and in sequence?
(Format: two words)

Q9. What is a collection of interconnected devices that communicate and share resources called?

Q10. What do you call the individual devices connected to a network?

Practice Questions — Set 3

Q11. What is the largest Wide Area Network that connects millions of LANs globally?

Q12. What is the acronym for a network that connects devices over a short distance (home, school, office)?

Q13. What term describes the communication pathways (wired or wireless) connecting nodes?
(Format: one word)

Q14. Which protocol provides FAST, connectionless communication at the Transport Layer?

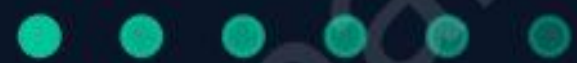
Q15. What device acts as a bridge between your home LAN and the ISP's WAN?

Quick Revision Cheat Sheet

Network	Interconnected devices sharing data	OSI L3	Network — IP, routers
Node	Individual device on a network	OSI L4	Transport — TCP/UDP
LAN	Local (home/office) — fast, small	OSI L5	Session — sessions, APIs
WAN	Wide (internet) — large, slower	OSI L6	Presentation — encrypt/decrypt
OSI L1	Physical — cables, raw bits	OSI L7	Application — HTTP, FTP, SMTP
OSI L2	Data Link — MAC, switches	TCP/IP	4 layers: Link → Internet → Transport → Application

NETWORK FOUNDATIONS

A complete guide from basics to advanced concepts



HTB Academy | Network Foundations

WHAT WE'LL COVER

01

Components of a Network

End devices, intermediary devices, servers & media

02

MAC Addresses

Hardware IDs, ARP, local communication

03

IP Addresses

IPv4 vs IPv6, logical addressing

04

Ports & Protocols

Well-known ports, TCP/UDP, HTTP/HTTPS

05

How Browsing Works

DNS, encapsulation, routing end-to-end

06

Practice Questions

Test your knowledge

COMPONENTS OF A NETWORK

End Devices	Intermediary Devices	Network Media	Servers
Computers & Laptops	Routers (Layer 3)	Ethernet cables	Web / File / Mail servers
Smartphones & Tablets	Switches (Layer 2)	Fiber-optic cables	Database servers
Smart TVs & IoT devices	Modems & Access Points	Wi-Fi & Bluetooth	Authentication servers
→ SOURCE & DESTINATION of data	→ DIRECT & FORWARD data	→ PHYSICAL PATHWAY for data	→ PROVIDE services to clients

END DEVICES & SERVERS — Deep Dive

END DEVICES (Hosts)

- ▶ Also called HOSTS — send & receive data
- ▶ Primary user interface to the internet
- ▶ Connect via Ethernet (wired) or Wi-Fi (wireless)
- ▶ Example: Student laptop on school Wi-Fi accessing e-learning portal
- ▶ Types: PCs, Macs, Phones, Smart TVs, IoT sensors

↔
response

SERVERS — Backbone of Internet

- ▶ Powerful computers providing services to CLIENTS
- ▶ Client-Server Model: server waits → client requests
- ▶ Web Server: hosts websites (apache, nginx)
- ▶ File Server: shared storage across a network
- ▶ Mail & DB Servers: email + structured data storage

ROUTERS & SWITCHES — Intermediary Devices



ROUTER

OSI Layer:	Network Layer — Layer 3
Function:	Forwards packets BETWEEN networks
Uses:	IP addresses to route data
Protocols:	OSPF, BGP for path selection
Security:	Built-in firewall, ACLs
Real Life:	Your home router connects all devices to ISP internet



SWITCH

OSI Layer:	Data Link Layer — Layer 2
Function:	Connects devices WITHIN a network (LAN)
Uses:	MAC addresses to forward frames
vs Hub:	Smart: sends only to intended recipient
Benefit:	Reduces congestion, improves performance
Real Life:	Office switch connects all PCs for file & printer sharing

MAC ADDRESSES

Media Access Control — Hardware Identity

00 : 1A : 2B : 3C : 4D : 5E

← OUI (Manufacturer — 24 bits) →

← Device ID (Unique — 24 bits) →

48-bit address

6 pairs of hex digits (e.g., 00:1A:2B:3C:4D:5E)

ARP Protocol

Maps IP addresses → MAC addresses

Layer 2 — Data Link

Used WITHIN a local network (LAN)

Globally Unique

No two NICs share the same MAC address

IP ADDRESSES — Logical Network Identity

IPv4

192 . 168 . 1 . 1

- 32-bit address space
- ~4.3 billion unique addresses
- 4 decimal numbers (0-255)
- Format: XXX.XXX.XXX.XXX
- Nearing exhaustion → IPv6 needed

V
S

IPv6

2001:0db8:85a3::8a2e:370:7334

- 128-bit address space
- 340 undecillion unique addresses
- 8 groups of 4 hex digits
- Format: XXXX:XXXX:....:XXXX
- Solves IPv4 exhaustion problem

PORTS — Traffic Directors at Layer 4

0 – 1023

Well-Known Ports

Reserved for common protocols (IANA managed)

80 → HTTP

443 → HTTPS

21/20 → FTP

25 → SMTP

1024 – 49151

Registered Ports

Assigned to specific apps by IANA

1433 → MS SQL Server

3306 → MySQL

8080 → Alt HTTP

5432 → PostgreSQL

49152 – 65535

Dynamic / Ephemeral

Temporary ports for client sessions

Browser ↔ Server

Auto-assigned

Closed after session

Also: custom apps

HOW BROWSING WORKS

What really happens when you type google.com and press Enter



PROTOCOLS, NICs & NETWORK MEDIA

Key Network Protocols

TCP/IP	Foundation of all internet communication	SMTP	Email transmission
HTTP/HTTPS	Web traffic — port 80 (HTTP) / 443 (HTTPS)	ARP	Maps IP → MAC address on local network
FTP	File transfers — ports 20 & 21	DNS	Resolves domain names to IP addresses

NICs & MEDIA

NIC:	Hardware card in device; has unique MAC address
Wired NIC:	Ethernet card + RJ-45 connector
Wireless NIC:	Wi-Fi adapter using radio waves
Fiber Optic:	Long distance, minimal signal loss
Ethernet:	LAN high-speed — Cat5e, Cat6
Firewall:	Software (IPTables) or hardware security

PRACTICE QUESTIONS

Test your understanding — answers from the content covered

1

What type of network cable is used to transmit data over long distances with minimal signal loss?

2

Which protocol manages data routing and delivery across networks?

3

What software is used to oversee and administer network operations? (3 words)

4

What software is used to protect individual devices from unauthorized network access? (1 word)

5

What type of cable connects components within a LAN for high-speed data transfer?

6

Which device connects multiple networks and manages data traffic to optimize performance?

7

What protocol maps IP addresses to MAC addresses?

8

Which IP version uses 128-bit addressing?

9

At which layer of the OSI model do ports operate? (two words)

10

What is the designated port number for HTTP?

11

What is the first step in the process of a web browsing session? (two words)

KEY TAKEAWAYS



End Devices

Send/receive data; primary user interface (PCs, phones, IoT)



IP Address

IPv4 (32-bit) vs IPv6 (128-bit); logical, routable address



Intermediary

Routers (L3, IP-based) & Switches (L2, MAC-based) direct data



Ports

0-1023 well-known; 1024-49151 registered; 49152+ dynamic



MAC Address

48-bit hardware ID; used within LAN; resolved via ARP



Security

Software firewalls (IPTables) protect at device level

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Part 1 Module 1